



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 27 summarizes the current fairness state for upload 27. Accuracy changed to +0.787 and fairness score changed to +0.827. Strong fairness improvement with no measurable accuracy loss. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.811.

ACCURACY	FAIRNESS SCORE	COMBINED SCORE	DISPARATE IMPACT
0.787 ▲ +0.787	0.827 ▲ +0.827	0.811 ThresholdOptimizer_mitigated_threshold	0.645 Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
compas.csv	compas_dt_baseline.pkl	comparison

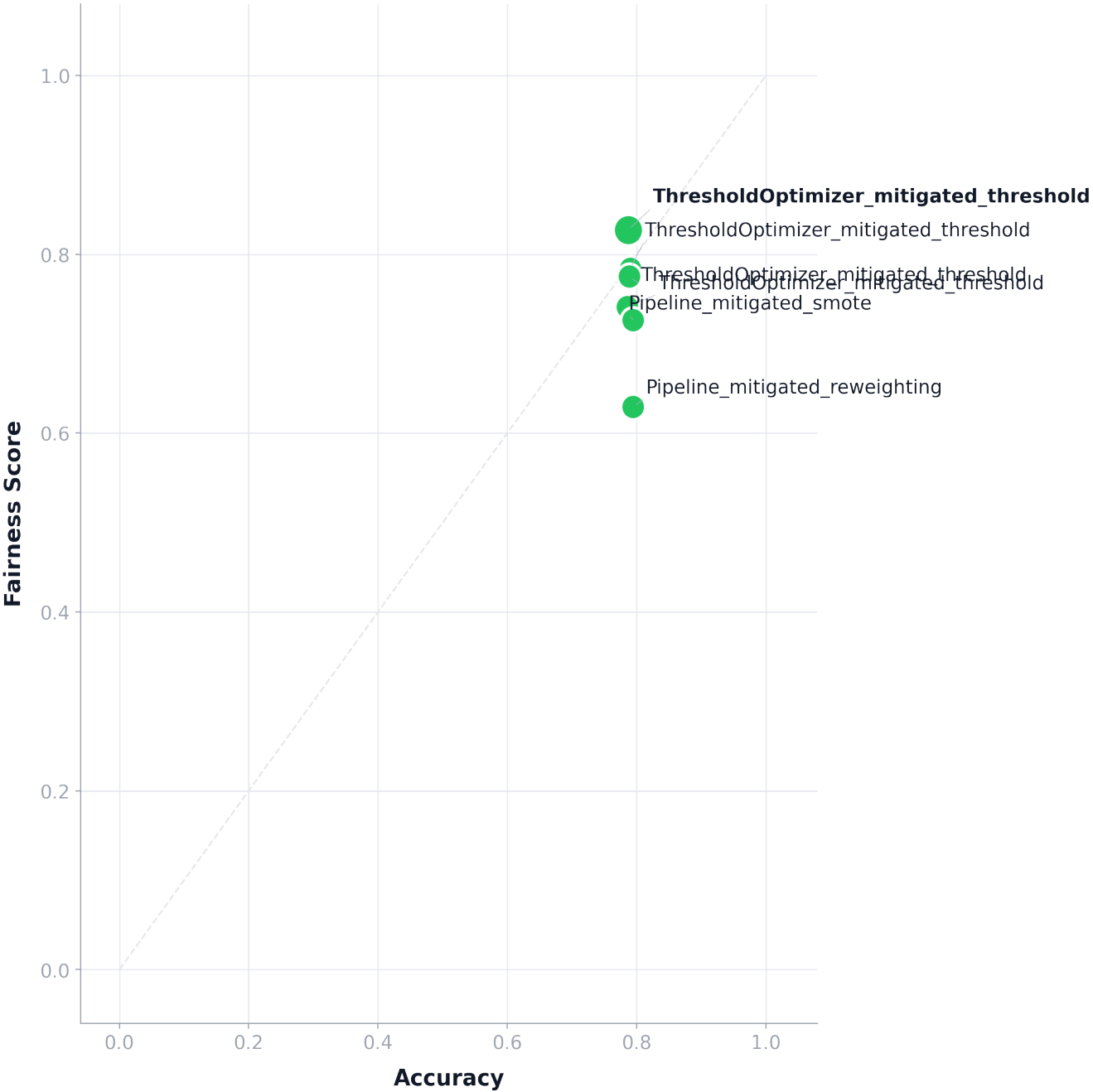
Fairness Metrics

METRIC	VALUE	INTERPRETATION
Demographic Parity Difference (DPD)	0.158	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.188	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.645	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.787	0.827	0.158	0.188	0.645	0.811
★ ThresholdOptimizer_mitigated_threshold	0.790	0.784	0.233	0.199	0.446	0.786
★ ThresholdOptimizer_mitigated_threshold	0.789	0.775	0.172	0.278	0.593	0.781
★ ThresholdOptimizer_mitigated_threshold	0.786	0.741	0.265	0.253	0.371	0.759
Pipeline_mitigated_smote	0.794	0.727	0.250	0.296	0.500	0.754
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Pipeline_mitigated_smote	0.794	0.727	0.250	0.296	0.500	0.754
Pipeline_mitigated_reweighting	0.794	0.629	0.337	0.405	0.394	0.695
Pipeline_mitigated_reweighting	0.794	0.629	0.337	0.405	0.394	0.695
Pipeline_mitigated_reweighting	0.794	0.629	0.337	0.405	0.394	0.695

Accuracy vs. Fairness Trade-off



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.